AELL101 LECTURE NOTES

COMPLETE RESPONSE AND AC CIRCUITS

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Group 10

COMPLETE RESPONSE:

DEFINITION:

The complete response of a circuit refers to the total response of a circuit variable (such as voltage or current) after a disturbance caused by switching or applying a source. It consists of two components: Natural response and Forced response.

- → The natural response is due to the stored energy in inductors and capacitors before the disturbance occurs.
- → A circuit's forced response is due to external independent sources (voltage or current) applied to it.

UNIT STEP FUNCTION:

The unit step function is a mathematical function used to represent signals that "turn on" at a specific time, usually at $t = 0$.

$$u(t) = \begin{cases} 0\,, & t < 0 \\ 1\,, & t \geq 0 \end{cases}$$

EXPRESSION:

- $v(t) = v_{transient} + v_{steady\ state}$
- $v_{transient}(t) = Ae^{-t/\tau}$
- $v_{steady\ state}(t) = V_s$
- $v(t) = Ae^{-t/\tau} + V_s$
- When the switch is closed at $t=t_0$,

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

$$i(t) = i(\infty) + [i(t_0) - i(\infty)]e^{-(t-t_0)/\tau}$$

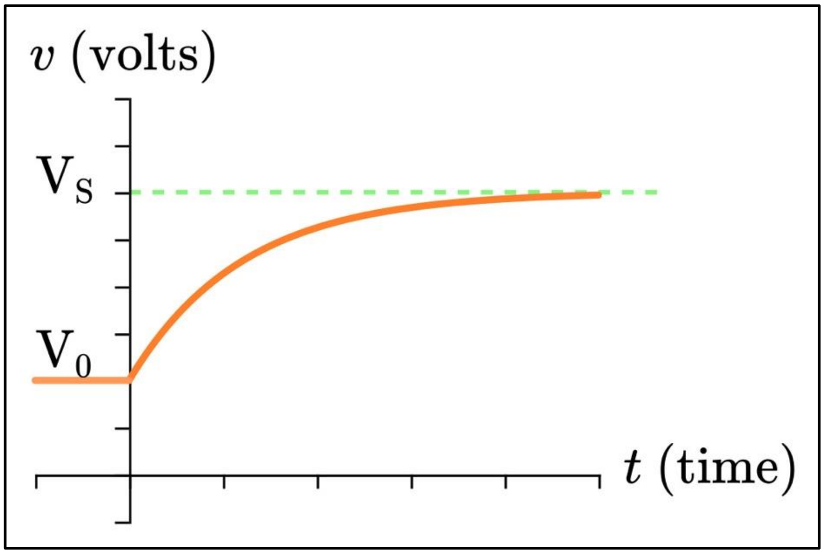
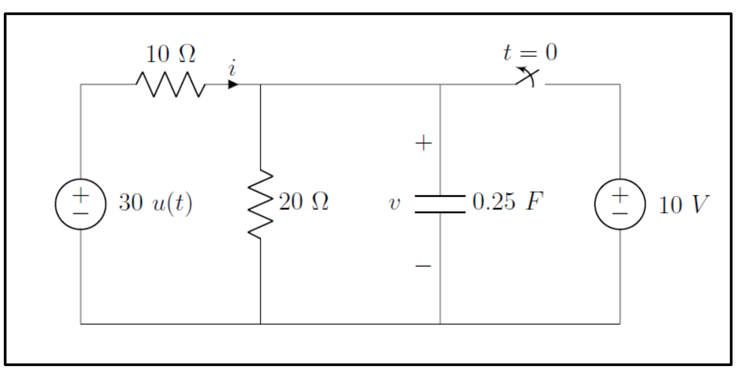


Fig.1: RC Step function graph

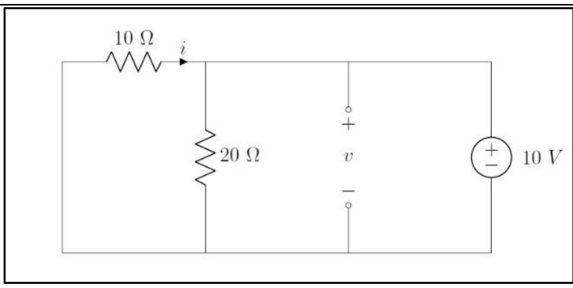
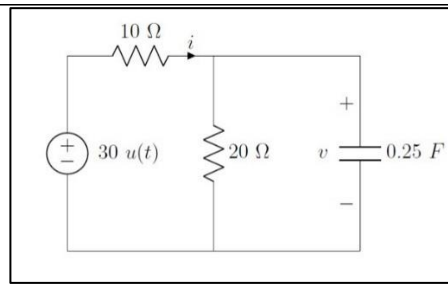
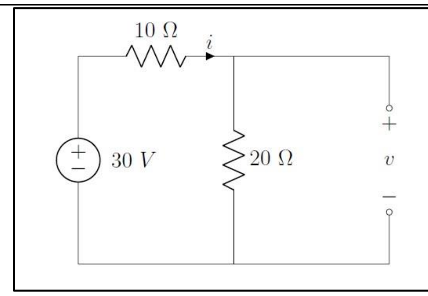
General procedure to find complete response:

1. Find the steady-state response [v(∞) or i(∞)]
2. Find the natural response without the independent source
3. Combine steps 1 and 2 to obtain the full expression
4. Apply initial conditions on the expression from 3 to find the unknown values

Question: Find v(t) for the given circuit, if switch is opened at t=0.



Answer:

At t = 0⁻:	***At t = 0⁺:***	***At t = ∞:***
		
$V_c(0^-) = 10\ V$ $R_{eq} = \frac{10\times20}{30} = \frac{20}{3}\ \Omega$	$V_c(0^+) = 10\ V$ $\tau = RC = \frac{20}{3} \times 0.25 = \frac{5}{3}\ s$ $V_{NS}(t) = 10e^{-3t/5}\ V$	$i = \frac{30}{30} = 1\ A$ $V_{SS} = V_{20\Omega} = 30 \times \frac{20}{(20+10)} = 20\ V$

$$\rightarrow v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

$$\therefore \boldsymbol{v(t) = 20 - 10e^{-3t/5}}$$

AC CIRCUITS:

PERIODIC WAVEFORMS:

Waveforms are graphical representations of periodic signals with the variation of a signal's amplitude over time by plotting each instantaneous value against the time axis.

→ These are also called <u>Alternating waveforms</u>.

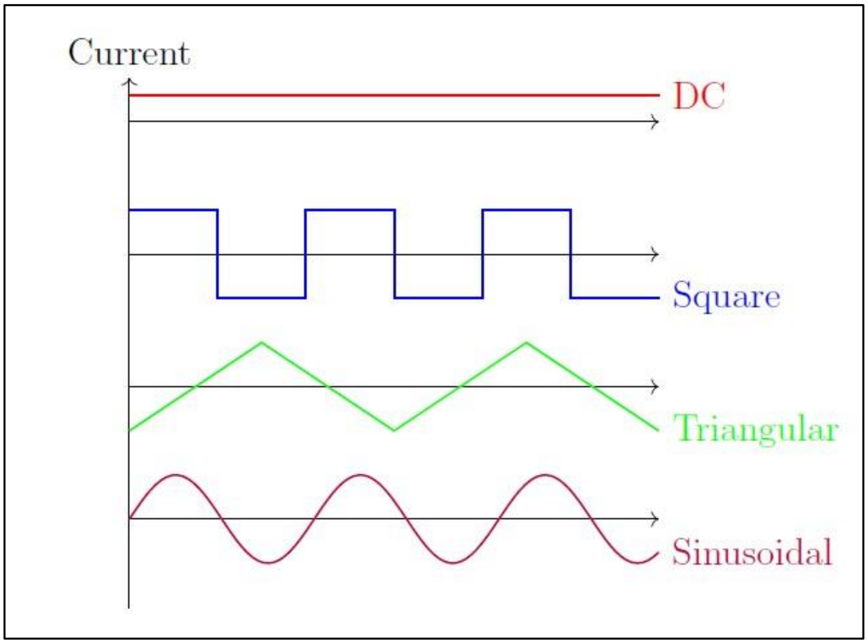


Fig.2: Types of waveforms

The sinusoidal wave is the most common of these waveforms, as it is also found abundantly in nature. For example, water waves resemble sinusoidal motion, sound waves propagate by a sinusoidal pattern, and AM and FM signals use sinusoidal waveforms for modulation.

KEY TERMS:

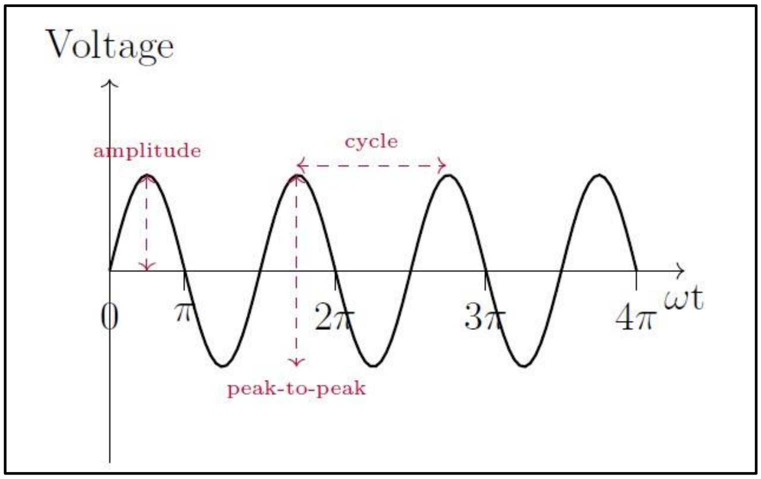


Fig 3: Components of an AC waveform

1) Waveform: An alternating quantity, when measured over a period of time, forms a characteristic shape known as its waveform.
2) Cycle: When the alternating quantity passes through 360°, it is called a cycle.
3) Time period (T): Time period is the time required to complete one cycle.
4) Frequency (f): Frequency refers to the number of cycles a quantity completes in one second.

$$f = \frac{1}{T}$$

5) Angular frequency (ω): The angular frequency of a sinusoidal wave represents the rate of angular displacement per unit.

$$\omega = 2\pi f = \frac{2\pi}{T}$$

6) Instantaneous value: The instantaneous value of a quantity is its value at a specific moment in time.
7) Peak-to-peak voltage: The peak-to-peak value is the total voltage change between the maximum and minimum points in one cycle of an alternating voltage or current.
8) Amplitude or Peak (V_m or I_m): The amplitude is the maximum peak value that voltage or current reaches during a complete cycle.

$$\boxed{\boldsymbol{v(t) = V_m \sin \omega t}}$$

PHASE:

- $v_1(t) = V_m \sin \omega t$
  $v_2(t) = V_m \sin \omega t + \phi$
- V_2 leads V_1 by ϕ (or) V_1 lags V_2 by ϕ.
- When $\phi = 0$:-
  V_1 and V_2 are in phase
- When $\phi \neq 0$:-
  V_1 and V_2 are out of phase

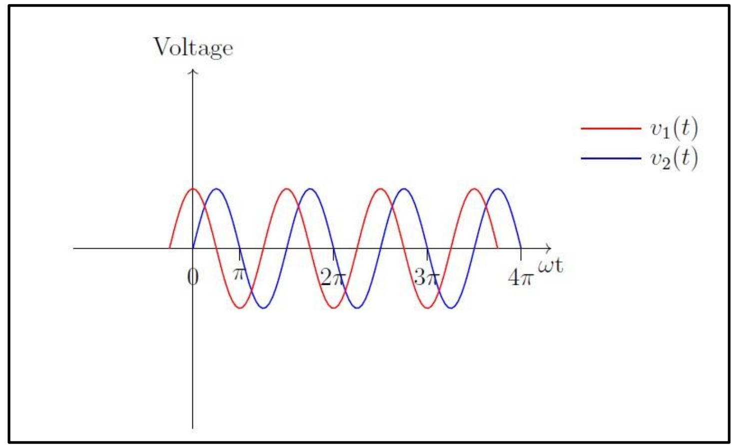


Fig 4: Two waveforms of different phase

IMPORTANT TRIGONOMETRIC IDENTITIES:

Taking $\omega t + \phi = x$ in radians,

- $\sin(x + 90°) = \cos x$
- $\sin(x - 90°) = -\cos x$
- $\cos(x + 90°) = -\sin x$
- $\cos(x - 90°) = \sin x$

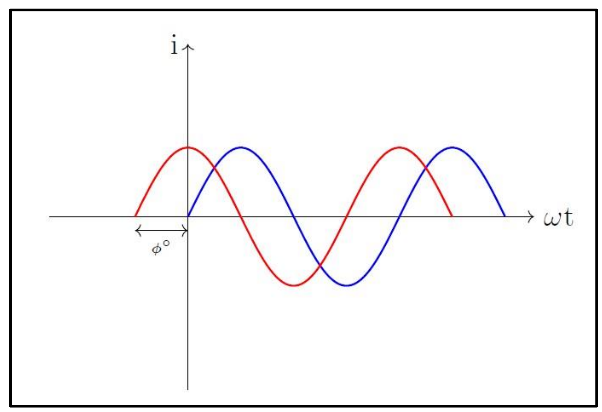


Fig 5: Two AC current waveforms

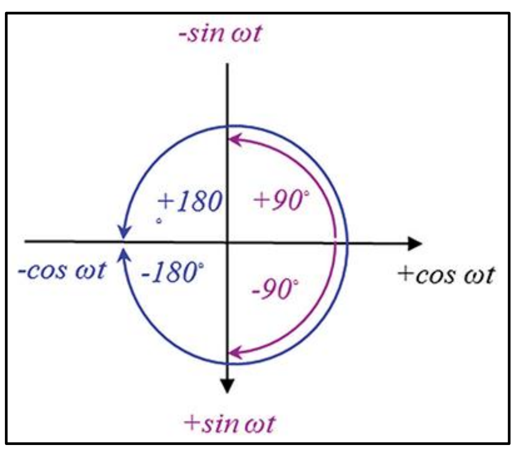


Fig 6: Interconversion between sine and cosine

EXAMPLES:

Example 1: $v(t) = 12cos(50t + 10°)$. Find the amplitude, phase, frequency, and time period of the sinusoid.

Answer: Comparing with $v(t) = V_m \, sin \, \omega t + \phi$,

- Amplitude: $V_{max} = 12$ V
- Angular Frequency: $v(t) = 12cos(50t + 10°) = 12 \sin(50t + 10° - 90°)$

  $\therefore v(t) = 12 \cos(50t + 10°) = 12cos(50t + 10°)$

  $\omega = 50 \, rad/s$
- Frequency: $f = \frac{1}{T} = \frac{\omega}{2\pi}$

  $f = \frac{50}{2\pi} = \frac{25}{\pi}$

  $f = 7.96 \, Hz$
- Phase: $\phi = -80°$
- Time Period: $T = \frac{2\pi}{\omega} = \frac{1}{f}$

  T =0.1257 s

Example 2: A sinusoidal current with 60 Hz frequency reaches the maximum value of 20A at t=2 ms. Write the equation of the current.

Answer: Since $i = i_0 \sin \omega t + \phi$,

$i_0 = 20 \, A$

$\omega = 2\pi f = 2\pi \times 60 = 377 \, rad/s$

At t=2 ms,

$20 = 20 \sin(377 \times 2 \times 10^{-3} + \phi) \Rightarrow \phi = 0.816 \, rad$

$\therefore i = 20 \sin(377t + 0.816)$ or $i = 20 \sin(377t + 46.8°)$

Example 3: $V_1 = -10\cos(\omega t + 50°)$ and $V_2 = 12\sin(\omega t - 10°)$. Which signal is leading and by how much?

Answer: To compare the two signals, convert V_1 to cos form to remove the negative sign and obtain a similar form as V_2.

$V_1 = -10\cos(\omega t + 50°)$

On using the identity $\sin(x - 90°) = -\cos x$,

$V_1 = 10\sin(\omega t + 50° - 90°) = 10\sin(\omega t - 40°)$

Hence, on comparing, V_2 leads V_1 by 30° (or) V_1 lags V_2 by 30°.

AVERAGE CURRENT (i_{avg}):

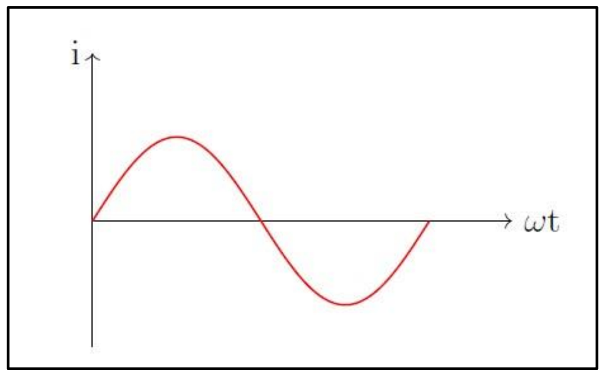


Fig 7: One cycle in an AC waveform

- $i = I_m \sin \omega t$
- For one cycle,

  $i_{avg,cycle} = 0$ A
- For a half-cycle,

$$i_{avg} = \frac{1}{T/2}\int_0^{T/2} i(t)dt$$

$$= \frac{2}{T}\int_0^{T/2} I_m \sin(\omega t)\, dt$$

$$= \frac{2I_m}{T}\left(\frac{-\cos\omega t}{\omega}\right)_0^{T/2}$$

$$= \frac{2I_m}{T\omega}\left[\cos 0 - \cos\left(\frac{2\pi}{T}\cdot\frac{T}{2}\right)\right] = \frac{2I_m}{T\cdot\frac{2\pi}{T}}\cdot 2$$

$$\therefore i_{avg,half\ cycle} = \frac{2I_m}{\pi}$$

Similarly,

$$v_{avg} = \frac{2V_m}{\pi}$$

AVERAGE POWER (P_{avg}):

$$P_{avg} = \frac{1}{T}\int_0^T i^2(t)R\,dt$$

$$= \frac{1}{T}\int_0^T {I_m}^2 \sin^2(\omega t)\,R\,dt$$

$$= \frac{{I_m}^2 R}{T}\int_0^T \frac{1-\cos 2\omega t}{2}\,dt$$, using the identity $\cos 2x = 1 - 2\sin^2 x$

$$= \frac{{I_m}^2 R}{2T}[T - \sin(2\omega t)]_0^T$$

$$= \frac{{I_m}^2 R}{2T}[T - (\sin\left(2 \cdot \frac{2\pi}{T} \cdot T\right) - \sin 0)]$$

$$\therefore P_{avg} = \frac{{I_m}^2 R}{2} = (\frac{I_m}{\sqrt{2}})^2 R = {I_{rms}}^2 R$$

Note:-

$$\boxed{I_{rms} = \frac{I_m}{\sqrt{2}}\,;\ V_{rms} = \frac{V_m}{\sqrt{2}}}$$

Example 4: Find V_{rms} for $V = 141.4\sin 377t$

Answer: $V_{rms} = \frac{141.4}{1.414} = 100\ V$

PHASOR:

- $e^{jx} = \cos x + j \sin x$ (Euler's identity)
- This can be related to v(t) in the following way:-

  $v(t) = V_m \cos(\omega t + \phi) \Leftrightarrow \frac{V_m}{\sqrt{2}} = \angle\phi°$

$$z = x + \iota y$$

$$x = r\cos\theta \,;\, y = r\sin\theta$$

$$r = \sqrt{x^2 + y^2} \text{ and } \theta = \tan^{-1}\frac{y}{x}$$

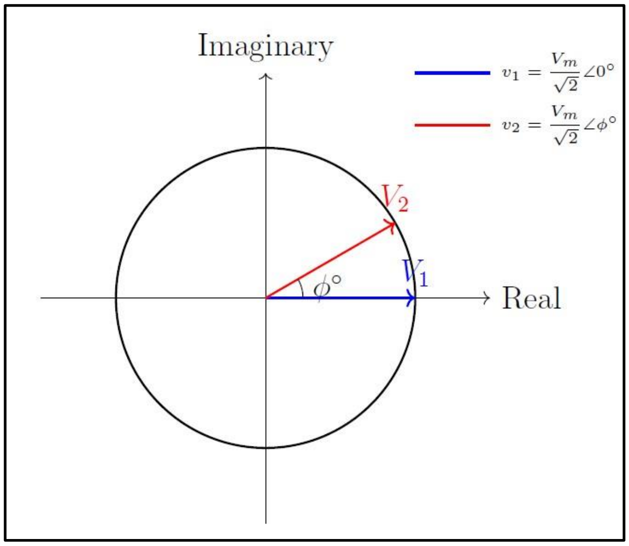


Fig 8: Phasor diagram for 2 voltages

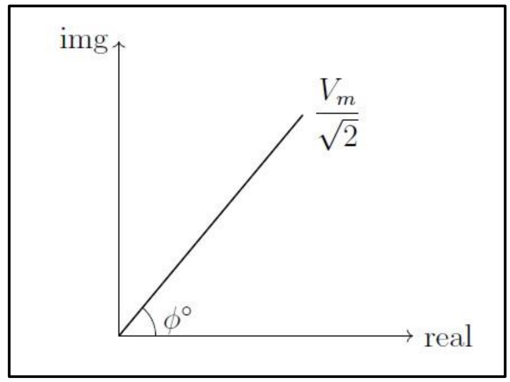


Fig 9: Graphical representation of V_m in the complex plane

Example 5: $i = 6\cos(50t - 40°)$. Convert i to a phasor.

Answer: $i = \frac{6}{\sqrt{2}}\angle - 40°$

Example 6: $i = -4\sin(\omega t + 50°)$. Convert i to a phasor.

Answer: On converting i into cosine form using the identity $\cos(x + 90°) = -\sin x$,

$$i = 4\cos(\omega t + 50° + 90°)$$

$$\therefore i = \frac{4}{\sqrt{2}}\angle 140°$$

Example 7: Find the sinusoidal expression for the following phasors:-

a) $v = -25\angle 40°$

b) $i = -j(12 - 5j)$

Answer: a) $v(t) = -25\sqrt{2}\cos(\omega t + 40°)$

b) $i = -12j - 5 = -5 - 12j$

$r = \sqrt{12^2 + 5^2} = 13;\ \theta = \tan^{-1}\frac{12}{5} = 67.38°$

$\therefore i = 13\sqrt{2}\cos(\omega t + 67.38°)$

AC IMPEDENCE:

- $i = I_m \cos\omega t = \frac{I_m}{\sqrt{2}}\angle 0°$
- $v = L\frac{di}{dt}$

  $= -\omega L I_m \sin\omega t$

  $= \omega L I_m \cos(\omega t + 90°)$

  $\therefore v = \frac{\omega L I_m}{\sqrt{2}}\angle 90°$
- $z = \frac{v}{i} = \omega L\angle 90°$

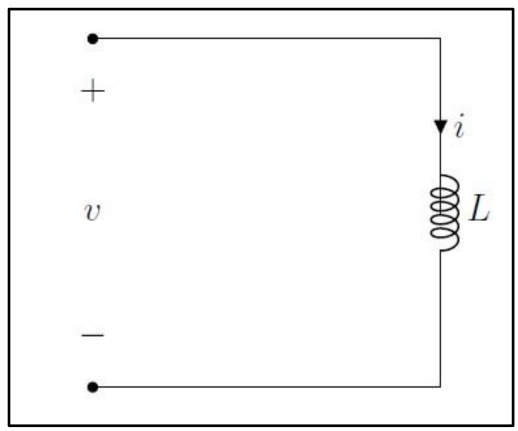


Fig 10: AC circuit with inductor

$$z_L = \omega L\angle 90°$$

$$z_C = \frac{1}{\omega C}\angle -90°$$

CONTRIBUTION STATEMENT:

- Noura Mohsin – Taking notes for lectures 1, 2, 3
- Shama – Taking notes for lectures 1, 2, 3
- Sunaina – Circuit diagrams for lectures 1, 2, 3 and compilation of notes
- Tarunika – Compilation of notes